

The background features a dark blue gradient with a subtle pattern of white stars and dots. Overlaid on this are several faint, light blue technical diagrams. On the left, there is a large circular scale with degree markings from 40 to 260 in increments of 10. Several concentric circles and arcs are drawn around this scale, some with arrows indicating a clockwise direction. Other smaller circular diagrams with arrows are scattered across the upper and lower portions of the image.

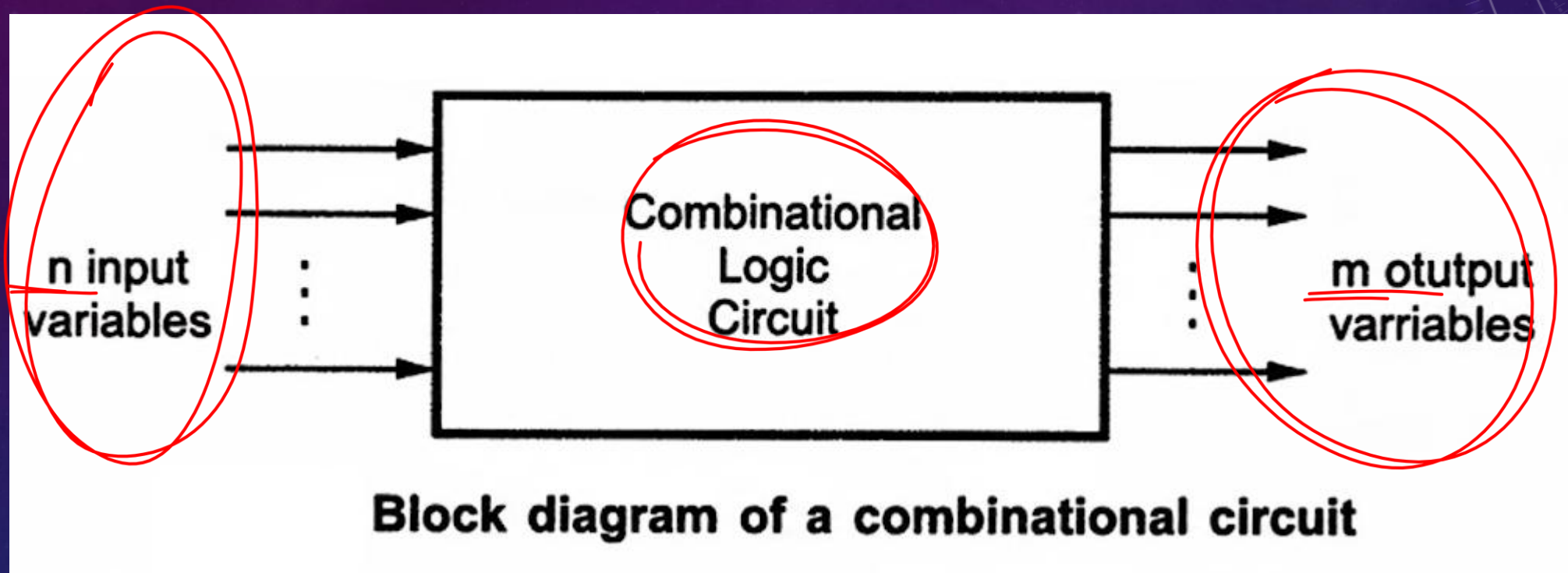
COMBINATIONAL LOGIC CIRCUITS

FUNDAMENTALS, DESIGN, AND IMPLEMENTATION

INTRODUCTION TO COMBINATIONAL LOGIC

- **Definition:** A digital logic circuit where the output depends only on the current combination of inputs.
- **Key Features:**
 - ✓ No memory (no storage elements).
 - ✓ No feedback loops.
 - ✓ Output changes as soon as inputs change (plus a tiny propagation delay).
- **Block Diagram Concept:** Show a box with n inputs and m outputs.

BLOCK DIAGRAM



THE DESIGN PROCEDURE

Explain the "Recipe" for creating any combinational circuit:

1. State the problem: What do we need to solve?
2. Determine variables: How many inputs (n) and outputs (m)?
3. Truth Table: List all 2^n combinations. *m terms / max terms*
4. Simplification: Use K-Maps or Boolean Algebra to find the simplest expression.
5. Logic Diagram: Draw the final gate-level circuit.

THREE WAYS OF SPECIFYING THE FUNCTION

- **Boolean Algebra** – This forms the algebraic expression showing the operation of the logic circuit for each input variable either True or False that results in a logic "1" output.
- **Truth Table** – A truth table defines the function of a logic gate by providing a concise list that shows all the output states in tabular form for each possible combination of input variable that the gate could encounter.
- **Logic Diagram** – This is a graphical representation of a logic circuit that shows the wiring and connections of each individual logic gate, represented by a specific graphical symbol, that implements the logic circuit.

Example

ARITHMETIC CIRCUITS - ADDERS

- Adders are the combinational circuits which is used to add two or more than two bits at a time.

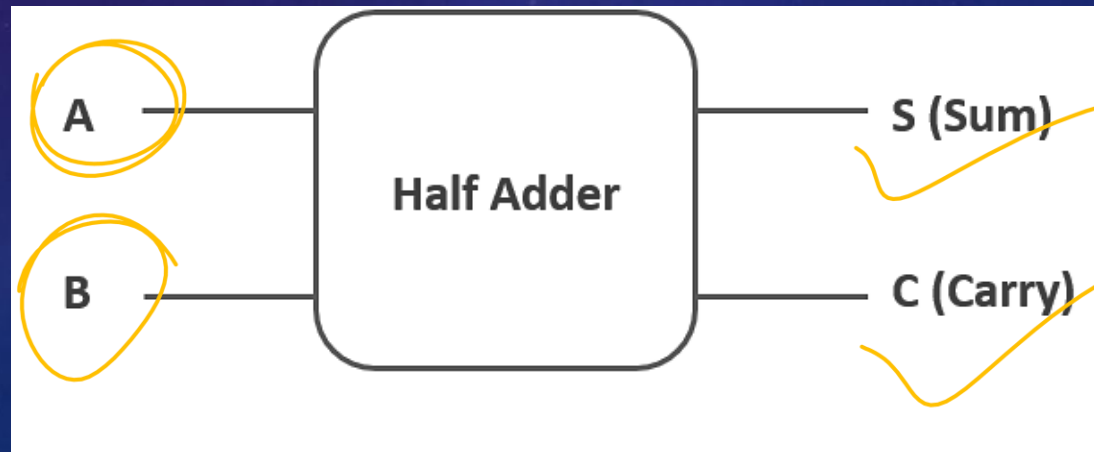
- Types of adders:

- Half Adder
- Full Adder

EX $a = 1$ $a = 1$
 $b = 0$ $b = 0$
 $\hline a + b = 1 + 0 = 1$ $c = 1$
 $\hline a + b + c = 1 + 0 + 1$

HALF ADDER:

- A combinational circuit that performs the addition of bits is called half adder.
- This circuit needs two binary inputs and two binary outputs.
- The input variables designate the augend(A) and addend(B) bits;
- The output variables produce the sum(S) and carry(C).
- The Half Adder is used to add only two 1-bit binary numbers



Handwritten calculations illustrating the addition of 1 and 0:

$$\begin{array}{r} 1 \\ + 0 \\ \hline 1 \end{array}$$

Below the addition, the result is written as $a+b=1+0=1$.

HALF ADDER TRUTH TABLE

A	B	Sum	Carry
0	0	0	0
0	1	1	0
1	0	1	0
1	1	0	1

min term
Index

m_0

m_1

m_2

m_3

SOP

$\text{Sum} = m_1 + m_2$

$\text{Carry} = m_3$

$A'B + AB'$

All the possible combinations of 1-bit binary additions are shown below:

1	1	0	0	← Input
+1	+0	+1	+0	← Input
10	1	1	0	
Carry		Sum		

FUNCTION SIMPLIFICATION

A	B	Sum	Carry
0	0	0	0
0	1	1	0
1	0	1	0
1	1	0	1

K-map:

For Carry

A \ B	0	1
0	0	0
1	0	1

For Sum

A \ B	0	1
0	0	1
1	1	0

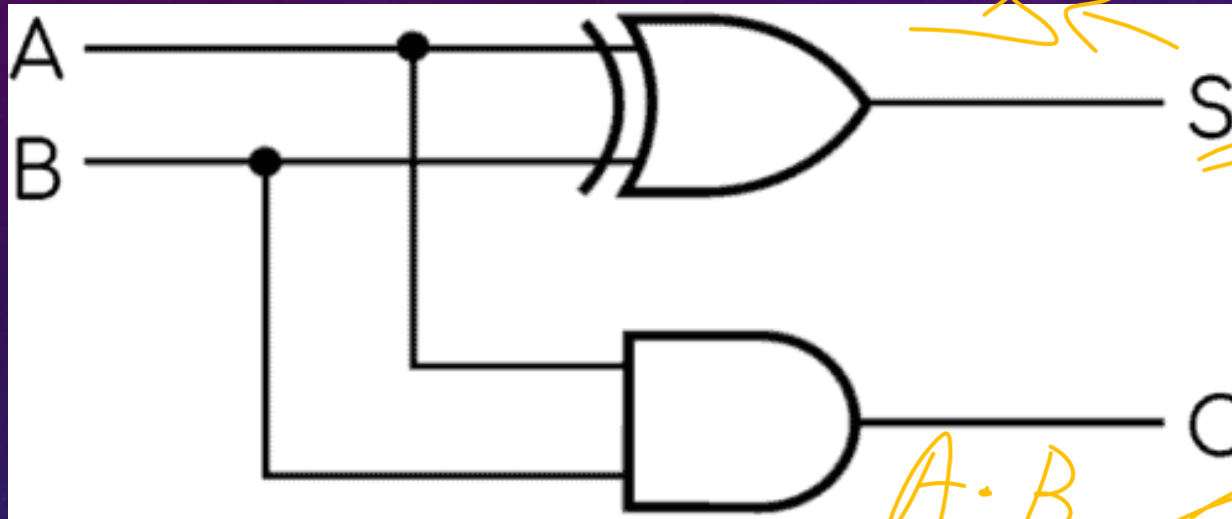
From k-map, the logical expression for:

• carry: $C = AB$

• Sum: $S = \bar{A}B + A\bar{B} = A \oplus B$



LOGIC DIAGRAM

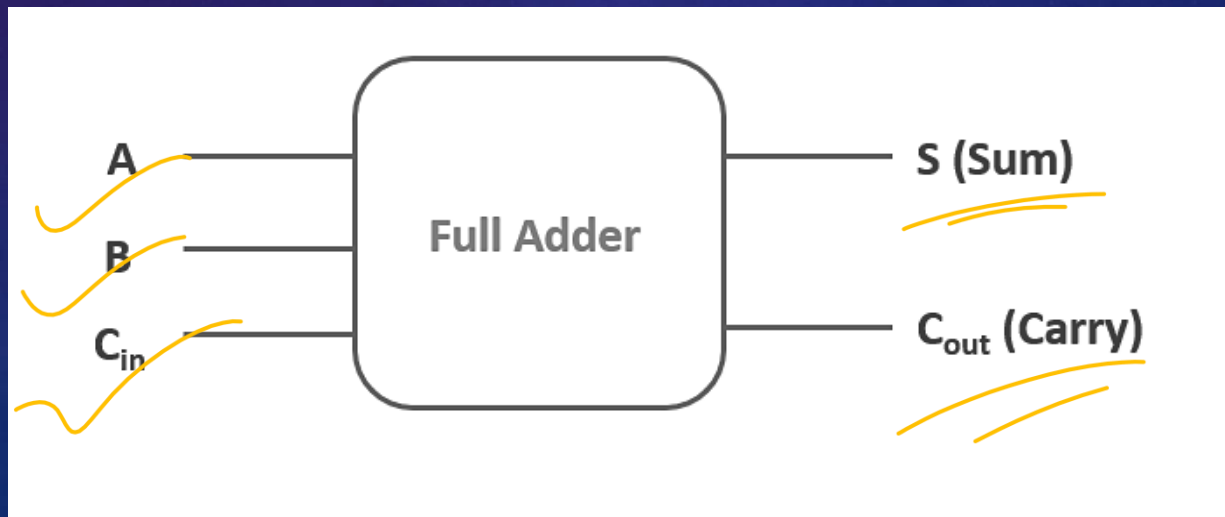


$A \oplus B$

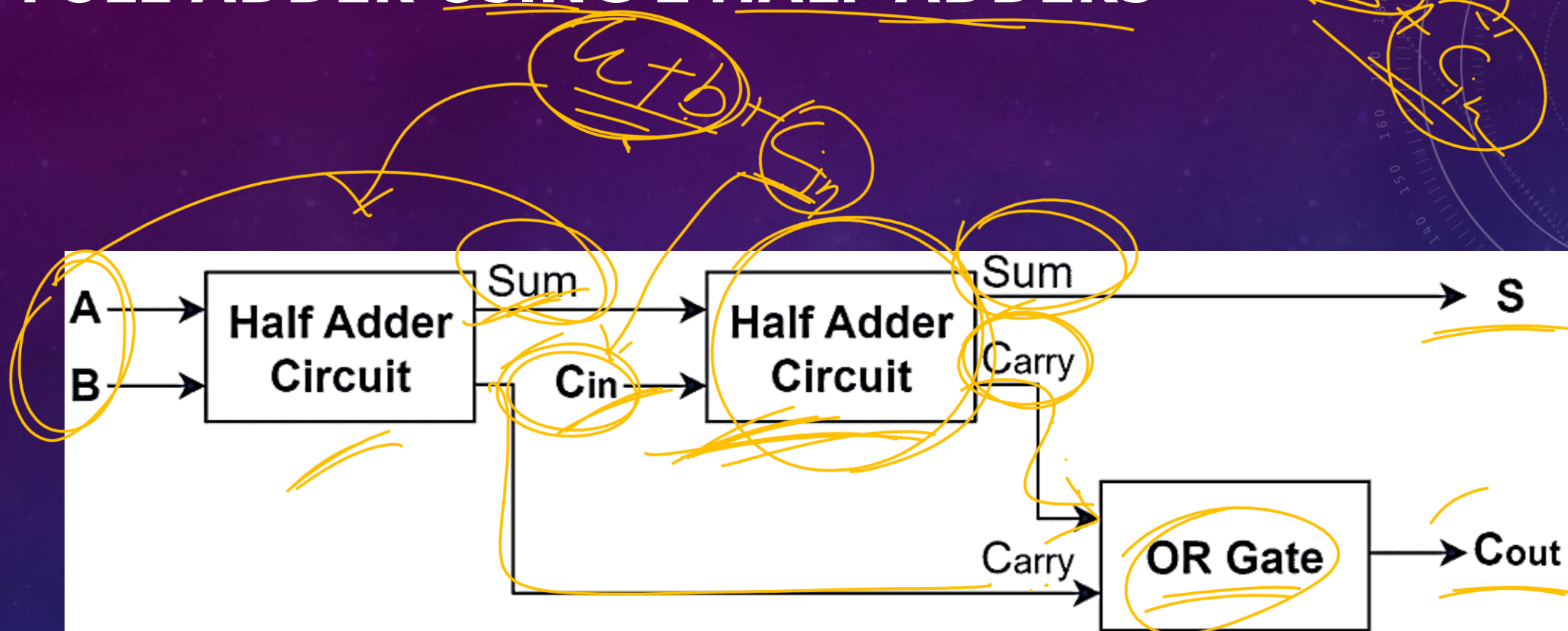
$A \cdot B$

FULL ADDER

- The full adder is the combinational circuit that adds the two bits along with the incoming carry (C_{in}).
- And generates the sum bit (S) and an outgoing carry bit (C_{out}) as an output.
- A Full Adder is able to add three 1-bit binary numbers.
- Meaning it can take a Carry_In bit from a previous adder.

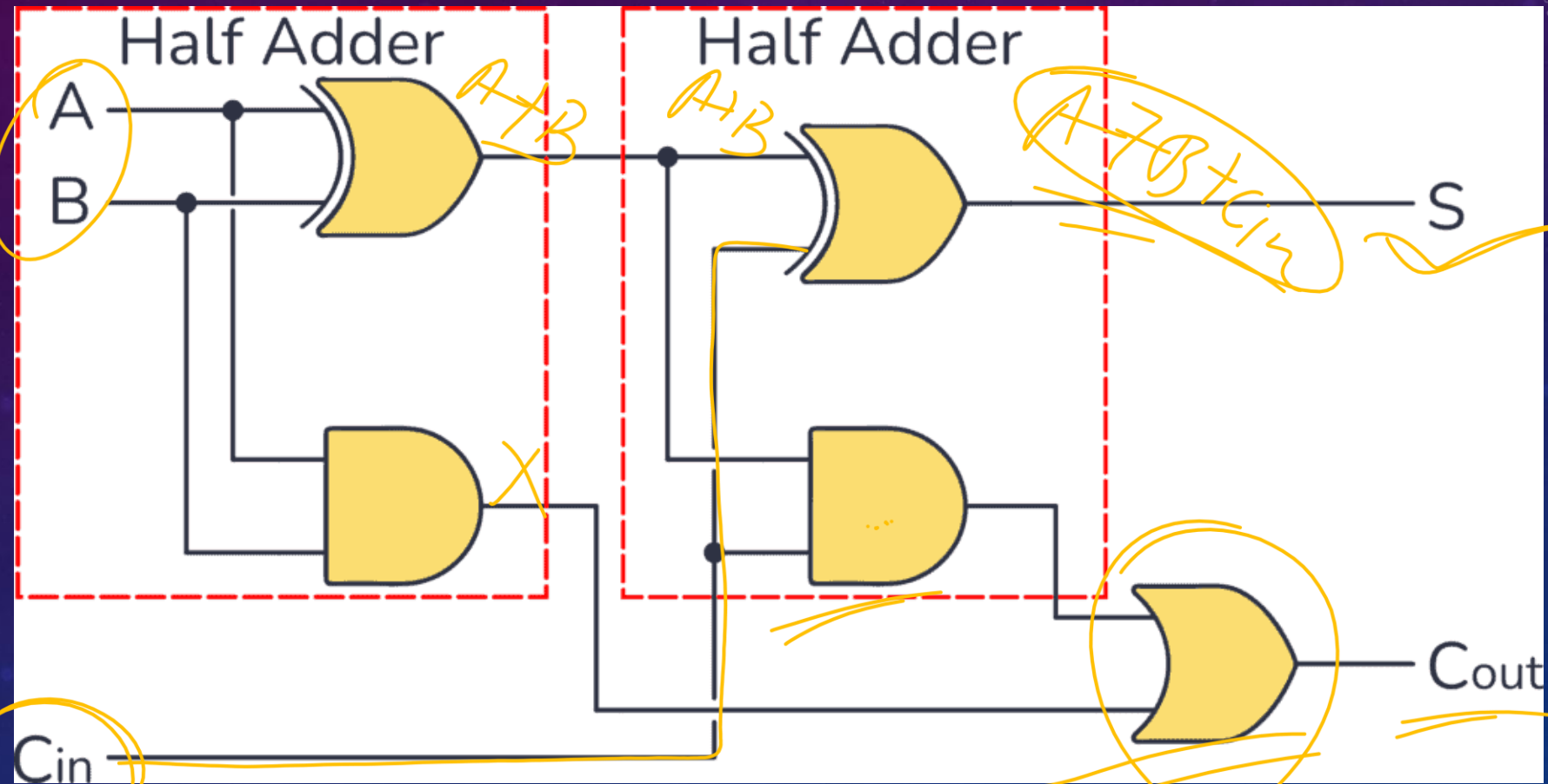


FULL ADDER USING 2 HALF ADDERS



$$a+b = \text{Sum} \\ \text{by } \underline{\text{Carry}}$$

LOGIC DIAGRAM

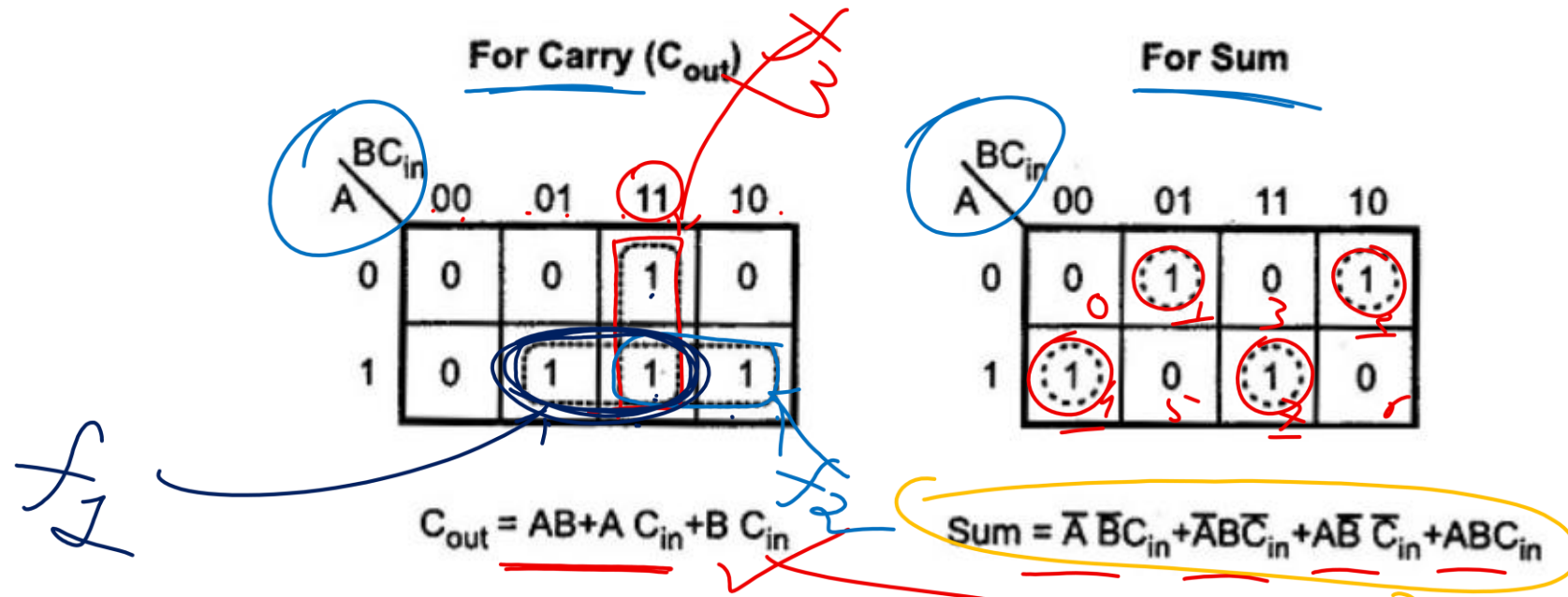


TRUTH TABLE OF THE FULL ADDER

<u>A</u>	<u>B</u>	<u>C_{in}</u>	Sum	C _{out}
0	0	0	0	0
0	0	1	1	0
0	1	0	1	0
0	1	1	0	1
1	0	0	1	0
1	0	1	0	1
1	1	0	0	1
1	1	1	1	1

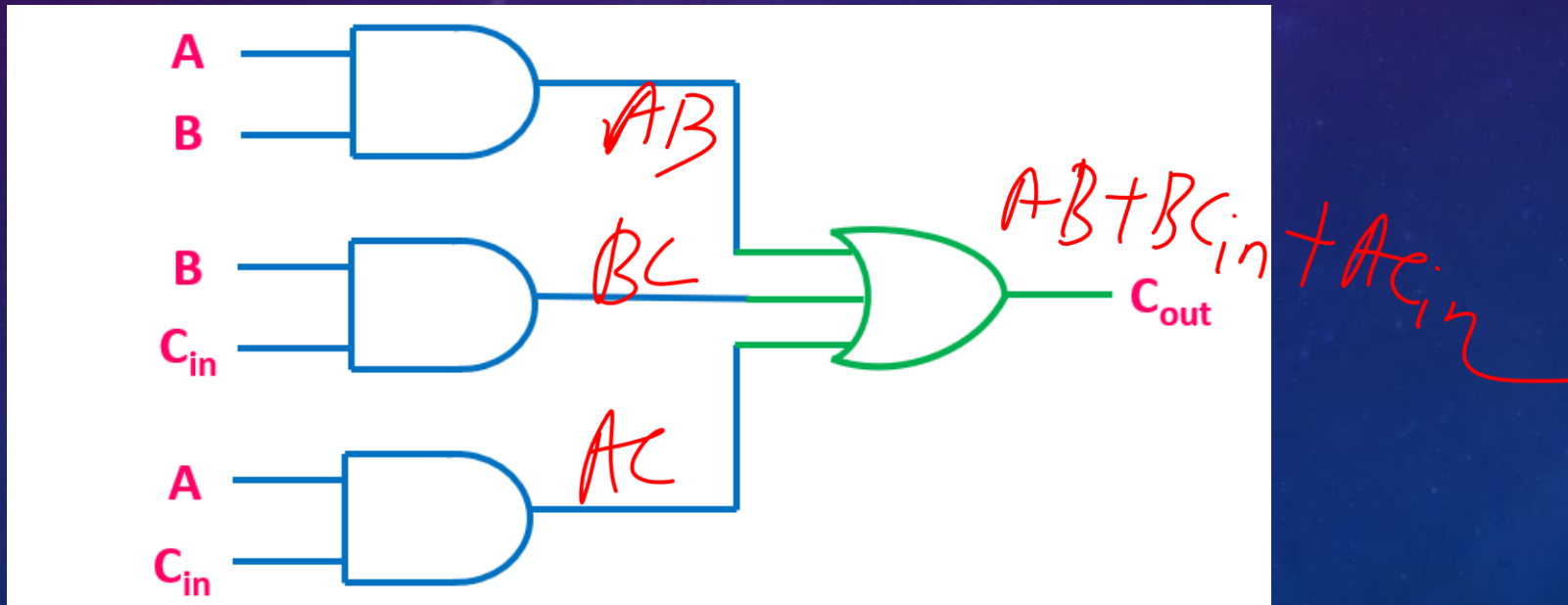
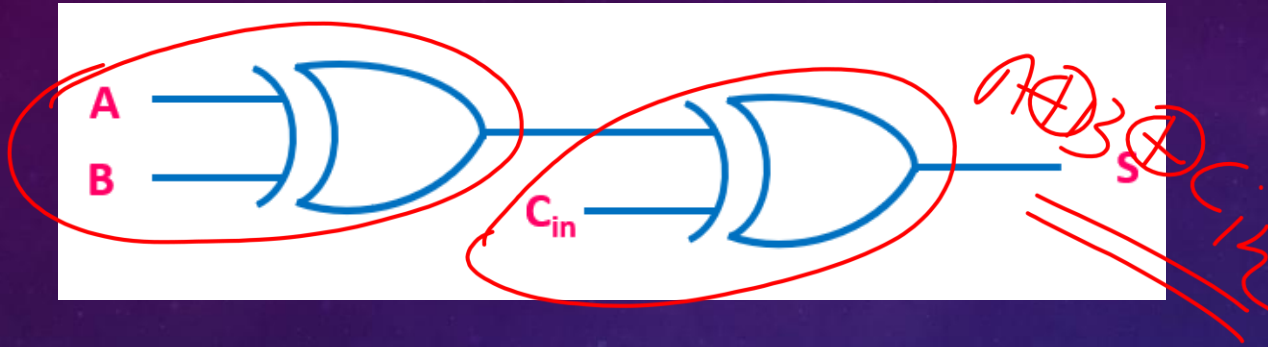
m
*m*₁
*m*₂
*m*₃
*m*₄
*m*₅
*m*₆
*m*₇

K-map simplification for carry and sum:

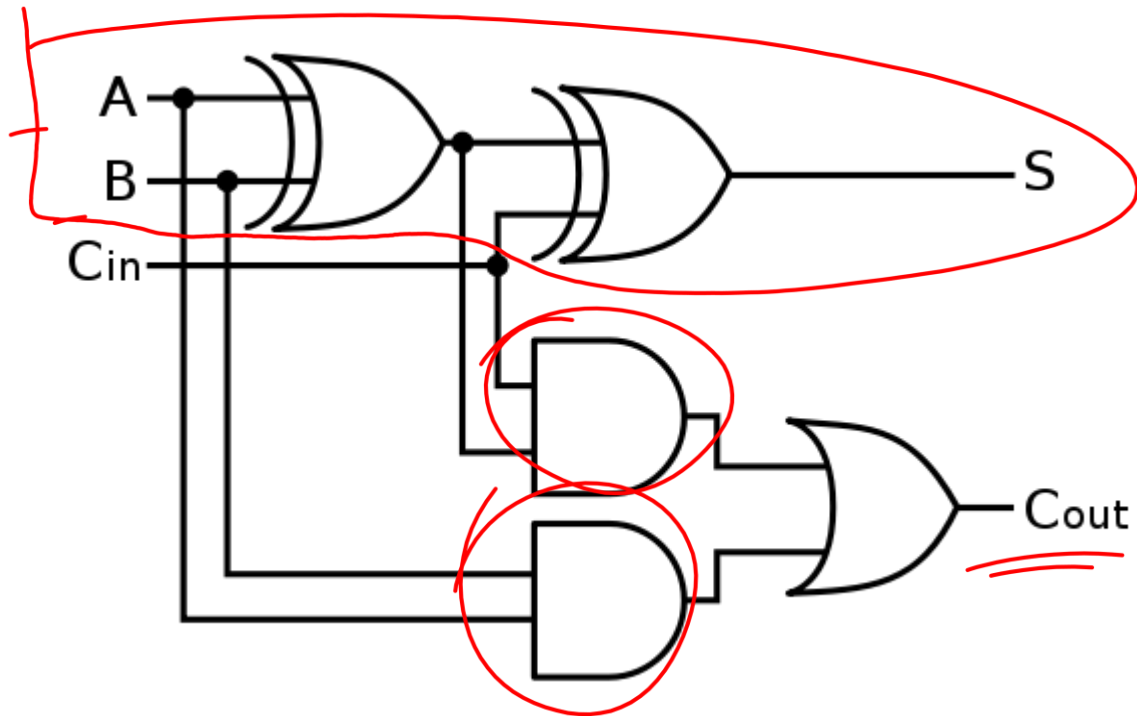


- The sum (S) of the full-adder is the XOR of A, B, and C_{in} .
- Therefore, Sum $S = A \oplus B \oplus C_{in} = \bar{A}\bar{B}C_{in} + \bar{A}B\bar{C}_{in} + A\bar{B}\bar{C}_{in} + ABC_{in}$
- The carry (C) of the half-adder is the AND of A and B.
- Therefore, Carry $C = \underline{AB} + \underline{AC_{in}} + \underline{BC_{in}}$

LOGIC DIAGRAMS:



COMBINED CIRCUIT DIAGRAM



THE MULTIPLEXER (MUX)

- A multiplexer is an electrical component capable of taking multiple inputs and giving a single output, depending on control or select inputs.
- In other words, we can use multiplexers to send multiple signals on the same line.
- for 2^n input lines, n selection lines are needed.
- MUX also known as Data Selector.

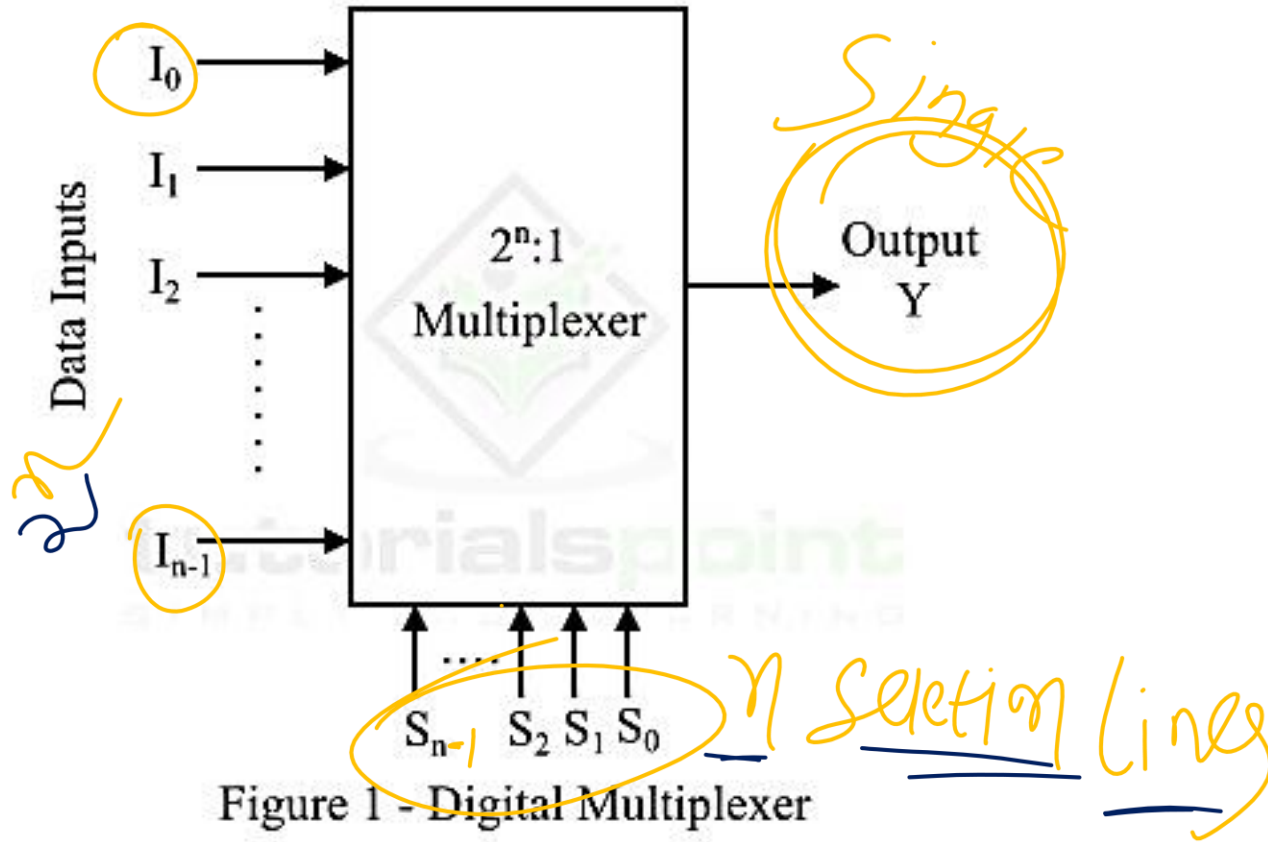
1/2/4/8/16. . .

2^n

Selection

I.V.

BLOCK DIAGRAM

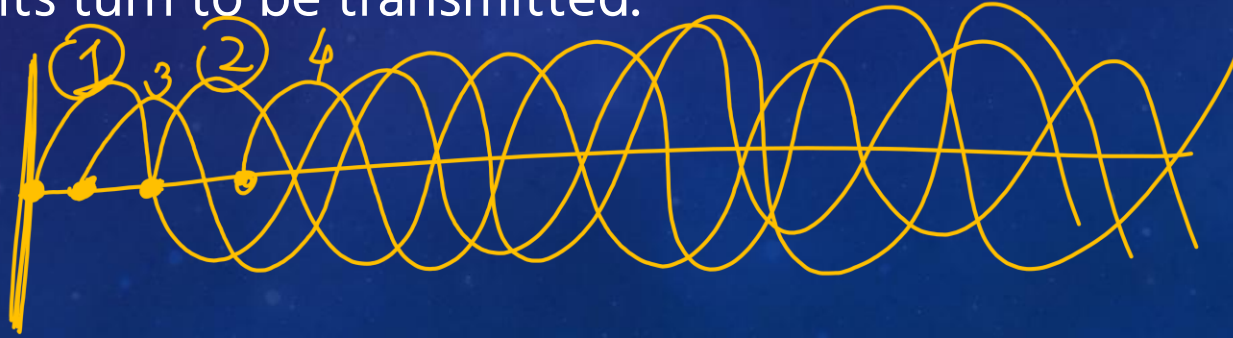


TYPES OF MULTIPLEXER (Domains)

1. Time division multiplexers

• When digital information is transmitted by a multiplexer time division multiplexers are commonly used.

- Multiple signals are carried over the same channel, but in different time slots, which cuts down on interference.
- To the user, transmission seems to be simultaneous, but each part of the signal actually waits for its turn to be transmitted.



2. Frequency division multiplexers

- When it comes to analogue communications, frequency division multiplexers are often used.
- Frequency division multiplexers enable pieces of data to be transmitted at different frequency levels, or sub-channels, simultaneously and in the same bandwidth.

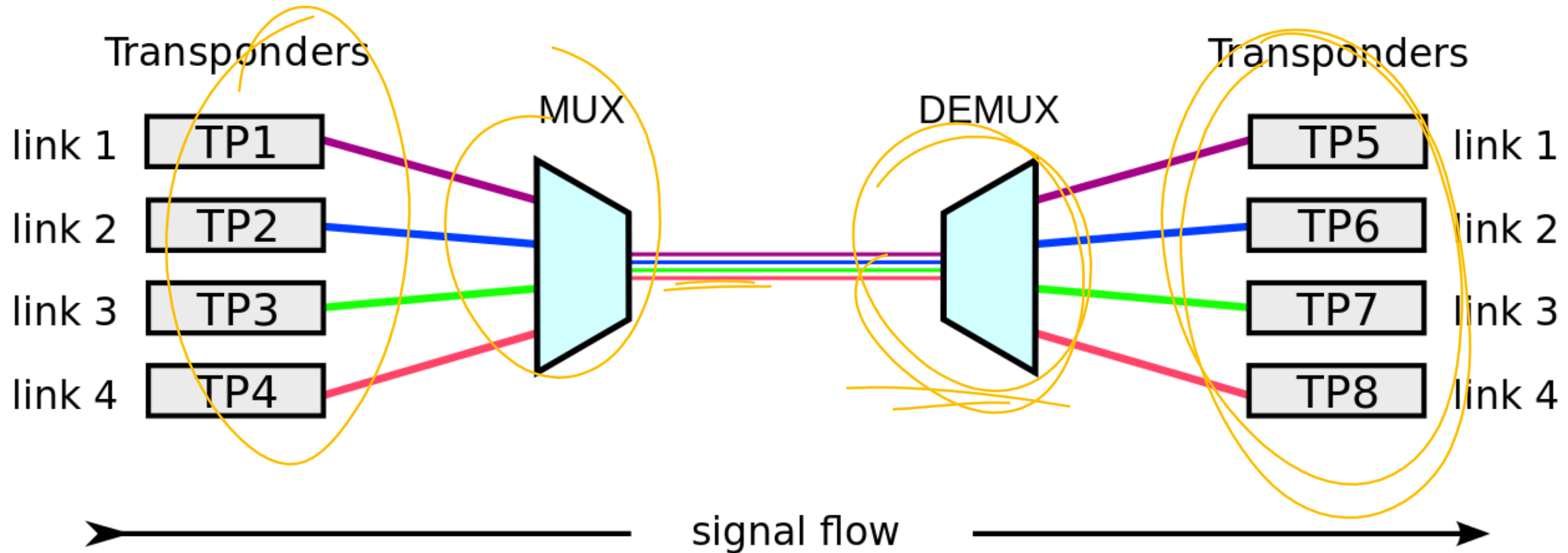




3. Wavelength division multiplexers

- Optical multiplexers and demultiplexers are used in fiber optic systems, and these include wavelength division multiplexer circuits.
- They use a shared channel – the fiber optic cable – but inside the cable each signal is transmitted on an individual light wavelength.
- Many wavelengths may be used to transmit signals simultaneously.
- It is also possible to transmit data in both directions, from the multiplexer circuit to the demultiplexer, and vice versa.

wavelength-division multiplexing (WDM)





fiber OPTIC

Light

4. Space division multiplexers

- Space division multiplexers use multiple channels.
- For example, it is possible to have a single fibre optic cable containing multiple optic fibres.
- Another option is to have one fibre containing multiple cores.
- Space division multiplexers are expected to help increase transmission capacity on networks.

High Quality
Communication

5. Statistical multiplexers

- Statistical multiplexers merge data from multiple input lines according to statistical laws.
- They provide a 'first come, first served' basis for merging data from multiple inputs and with variable bit rates into one.
- Statistical multiplexers make real-time transmission of different streams possible – without compromising the quality of audio or video information.

6. Inverse multiplexers

- What's the opposite of a multiplexer? The answer is an inverse multiplexer.
- These do the opposite of most multiplexer circuits because they break down one data stream into multiple parts. → Packets → frames
- Inverse multiplexers, however, differ from demultiplexers because these parts of data remain connected, rather than being unrelated.

eg: / 100!
↳ 100th = 1st

eg: / 100!
↳ 47th Rank

7. Code division multiplexers

- Code division multiplexers combine a number of data signals by giving each one a unique code so they can be transmitted simultaneously.
- Advantages of code division multiplexing include a signal that is less prone to interference and is more private.

Ex: VIP culture

highest Priority

TYPES OF MUX (BASED ON INPUT LINES)

2ⁿx1 Mux:

• 2x1 Mux

• 4x1 Mux

• 8x1 Mux

• 16x1 Mux

$\Rightarrow 2^1 \times 1$

$\Rightarrow 2^2 \times 1$

$\Rightarrow 2^3 \times 1$

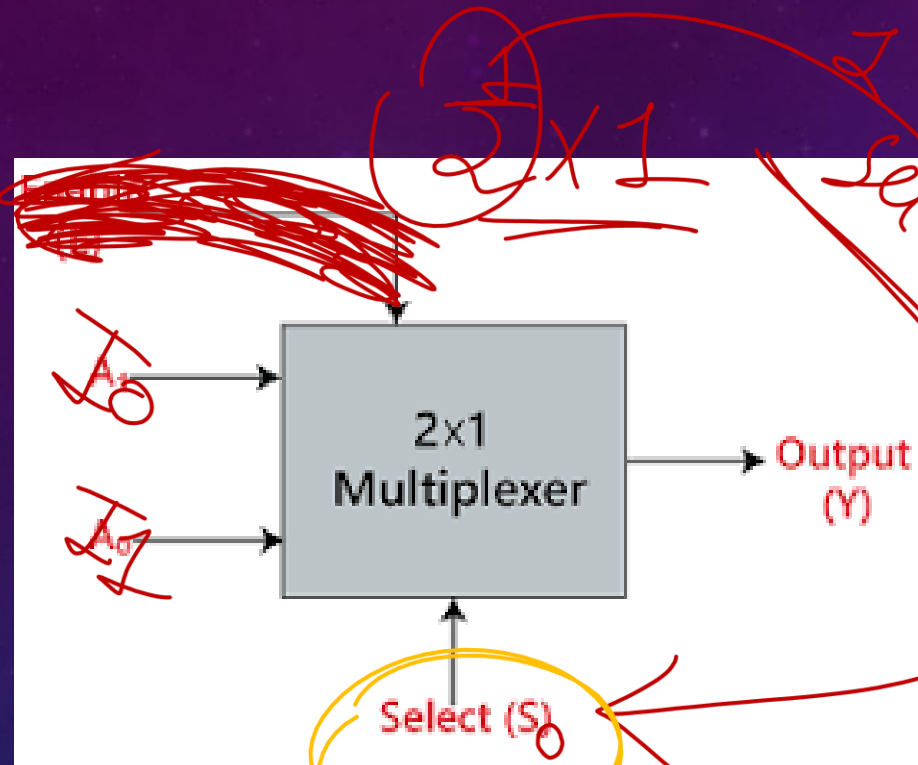
$\Rightarrow 2^4 \times 1$

Design

2X1 MULTIPLEXER

- The 2x1 is a fundamental circuit which is also known 2-to-1 multiplexer that are used to choose one signal from two inputs and transmits it to the output.
- The 2x1 mux has two input lines, one output line, and a single selection line.
- It has various applications in digital systems such as in microprocessor it is used to select between two different data sources or between two different instructions.

BLOCK DIAGRAM & TRUTH TABLE



S_0	I_0	I_1	Y
0	0	X	0
0	1	X	1
1	X	0	0
1	X	1	1

$S_0 \rightarrow 0 \Rightarrow O/P \Rightarrow I_0$
 $S_0 \rightarrow 1 \Rightarrow O/P \Rightarrow I_1$

WORKING

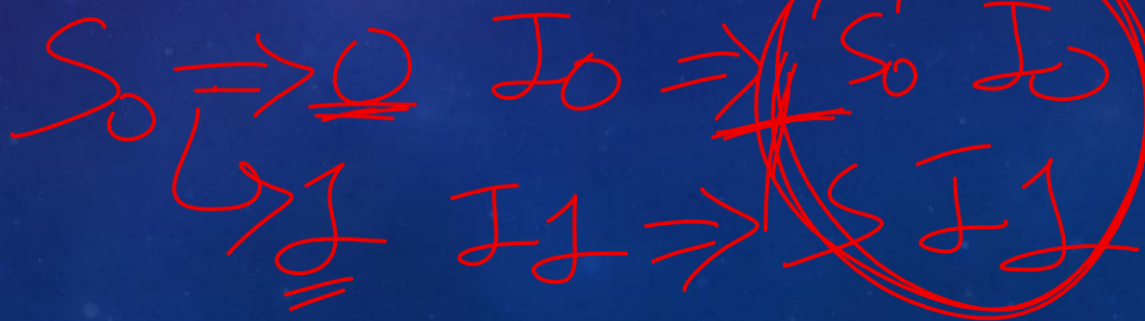
- The output of the 2x1 Mux will depend on the selection line S_0
- When S_0 is 0(low), the I_0 is selected $\Rightarrow S_0 \Rightarrow 0 \rightarrow \underline{O/P}$
- when S_0 is 1(High), I_1 is selected $\Rightarrow S_0 \Rightarrow 1 \rightarrow \underline{O/P}$



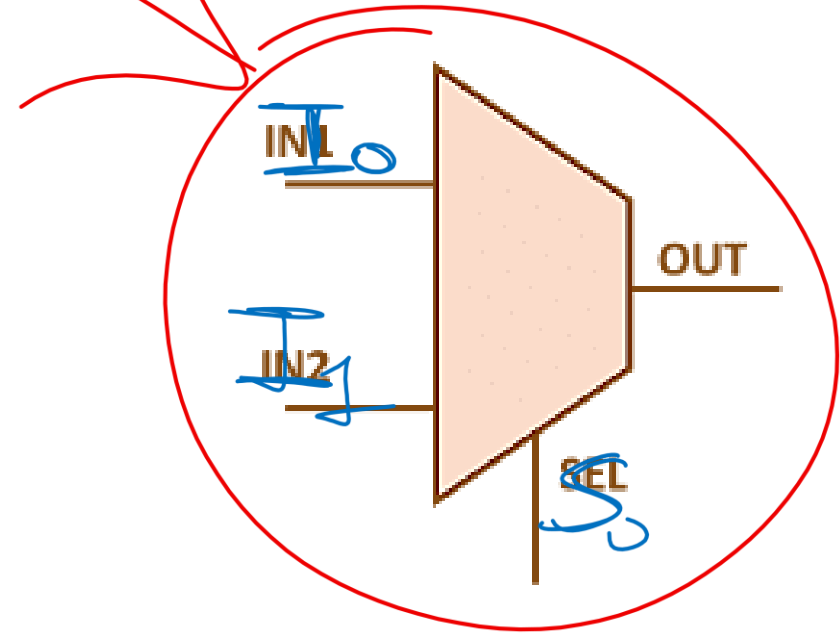
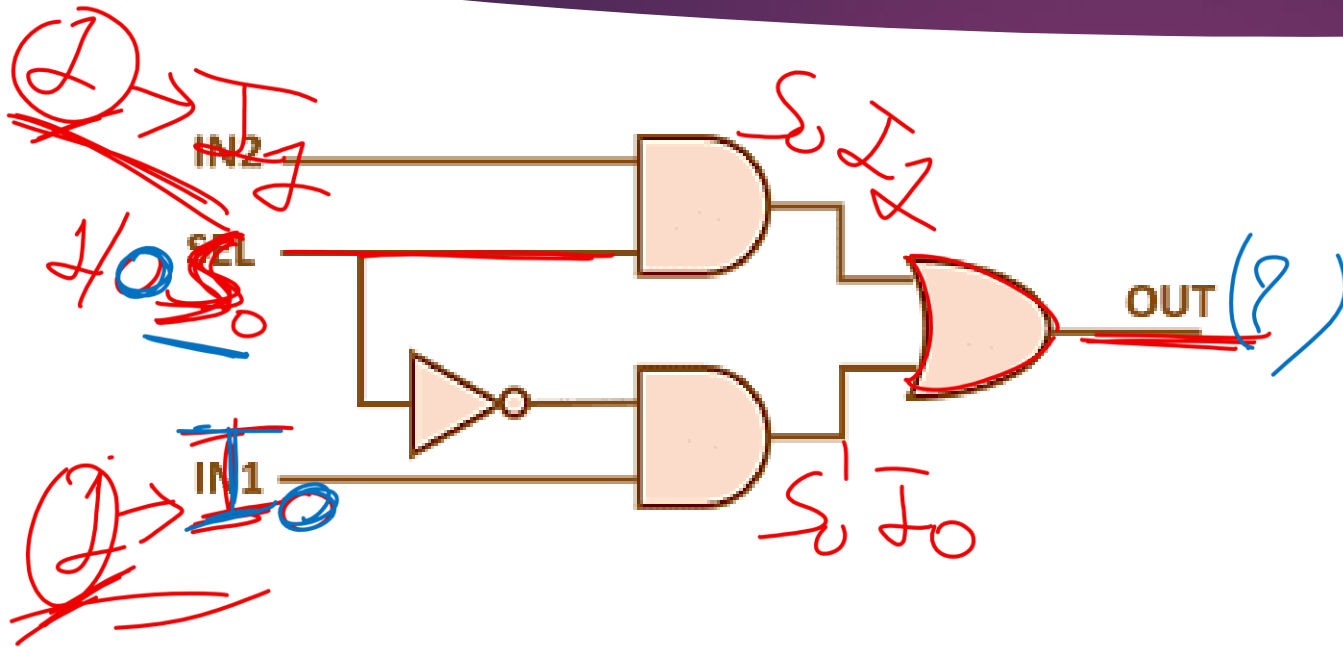
Logical Expression of 2x1 Mux:

- Using the Truth Table ,the Logical Expression for Mux can be determined as:

$$Y = S_0' \cdot I_0 + S_0 \cdot I_1$$

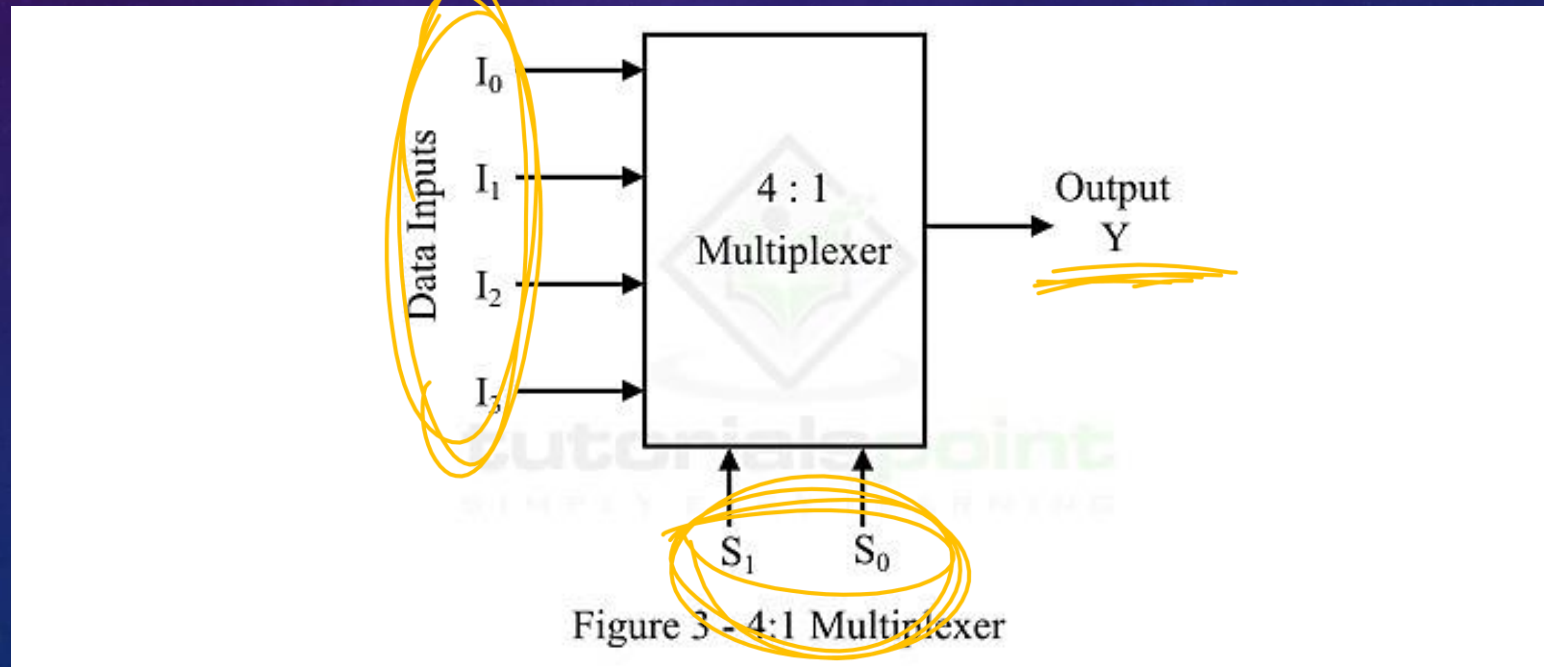


Logic Diagram & Symbol



4×1 MULTIPLEXER

- 4×1 Multiplexer has four data inputs I_3, I_2, I_1 & I_0 , two selection lines s_1 & s_0 and one output Y .
- The Output is selected as one of the 4 inputs which is based on the selection inputs.
- The block diagram of 4×1 Multiplexer is shown in the following figure.



WORKING

The output of the multiplexer is determined by the binary value of the selection lines

- When $S_1 S_0 = 00$, the input I_0 is selected.
- When $S_1 S_0 = 01$, the input I_1 is selected.
- When $S_1 S_0 = 10$, the input I_2 is selected.
- When $S_1 S_0 = 11$, the input I_3 is selected.

Handwritten notes:
I/P (circled)
O/P (circled)
A vertical line connects the circled I/P and O/P labels to the list of selection line combinations.

TRUTH TABLE OF 4×1 MULTIPLEXER

Selection Lines		Output
S_1	S_0	Y
<u>0</u>	<u>0</u>	<u>I_0</u>
<u>0</u>	<u>1</u>	<u>I_1</u>
<u>1</u>	<u>0</u>	<u>I_2</u>
<u>1</u>	<u>1</u>	<u>I_3</u>

From Truth table, we can directly write the Boolean function for output, Y as

$$Y = \underline{S_1'S_0'I_0} + \underline{S_1'S_0I_1} + \underline{S_1S_0'I_2} + \underline{S_1S_0I_3}$$

CIRCUIT DIAGRAM

